

Shaoheng Liang

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EXPERIENCE

- **Lane Fellow (Postdoctoral Researcher)** 2023 – Present
Carnegie Mellon University | Advisor: Jian Ma (Ray and Stephanie Lane Computational Biology Department) Pittsburgh, PA
- **Consultant, Bioinformatics** 2023
Torus Biosystems Medford, MA
- **Research Scientist, Bioinformatics** 2022 – 2023
NuProbe USA, Inc. Houston, TX
- **Graduate Researcher** 2017 – 2022
MD Anderson Cancer Center | Advisor: Ken Chen (Computational Cancer Genomics Lab) Houston, TX
- **Undergraduate Researcher** 2015 – 2016
Tsinghua University | Advisor: Jiansheng Chen (Image Recognition Lab) Beijing, China
- **Software Engineer Intern** 2014
Schneider-Electric China R&D Center Shanghai, China

EDUCATION

- **Ph.D. in Computer Science & Certificate in Teaching and Learning** 2017 – 2022
Rice University | Advisors: Ken Chen & Luay Nakhleh Houston, Texas
Dissertation: Interpretable and Efficient Machine Learning in Cancer Biology
- **Tsinghua University | Research Advisor: Jiansheng Chen** 2013 – 2017
B.Eng. in Electronic Information Science and Technology Beijing, China

HONORS AND AWARDS

- **Lane Fellowship** | Carnegie Mellon University 2022 – Present
- **Hongqian Electronics Scholarship for Scientific Innovation** | Tsinghua University 2016
- **University Scholarship for Academic Excellence** | Tsinghua University 2016
- **First Prize, National Olympiad in Informatics in Provinces** | China Computer Federation 2012
- **First Prize, National Olympiad in Informatics in Provinces** | China Computer Federation 2013

LEADERSHIP AND SERVICES

- **Session moderator** | Retreat, Ray and Stephanie Lane Computational Biology Department 2026
- **Judge** | Carnegie Mellon x NVIDIA Federated Learning Hackathon for Biomedical Applications 2025
- **Postdoc member** | Postdoc and Student Committee on Accessibility and Insitutional Support 2023
- **Mentor (career advice)** | CMU Undergraduate AI Mentoring Program 2023
- **Guest editor** | *Emerging Topics in Life Sciences on AI in the life sciences* 2021
- **Reviewer** | *Cell Systems, Nature Communications, BioData Mining, BMC Bioinformatics, PLOS Computational Biology, Scientific Reports, RECOMB {2020, 2024–2026}, AICoB 2019, and ISMCO 2020* 2020 – Present

Featured publications

- [1 S] Sizhe Liu, Junjie Tang, Jian Ma[†], and **Shaoheng Liang**[†]. “TissueNarrator: Generative Modeling of Spatial Transcriptomics with Large Language Models”. *BioRxiv* (2025), pp. 2025–11. (Under review at *Nature Biomedical Engineering*).
- [2 C] **Shaoheng Liang**, Junjie Tang, Guanghan Wang, and Jian Ma. “STEAMBOAT: Attention-based multiscale delineation of cellular interactions in tissues”. *International Conference on Research in Computational Molecular Biology (RECOMB)*. 2025, pp. 432–436. (Extended version under review at *Nature Methods*).
- [3 J] Jiakai Hou*, **Shaoheng Liang***, Chunyu Xu, Yanjun Wei, Yunfei Wang, Yukun Tan, Nidhi Sahni, Daniel J McGrail, Chantale Bernatchez, Michael Davies, et al. “Single-cell CRISPR immune screens reveal immunological roles of tumor intrinsic factors”. *NAR Cancer* 4.4 (2022), zcac038.
- [4 J] **Shaoheng Liang**, Vakul Mohanty, Jinzhuang Dou, Qi Miao, Yuefan Huang, Muharrem Müftüoğlu, Li Ding, Weiyei Peng, and Ken Chen. “Single-cell manifold-preserving feature selection for detecting rare cell populations”. *Nature Computational Science* 1.5 (2021), pp. 374–384.
- [5 J] **Shaoheng Liang**, Fang Wang, Jincheng Han, and Ken Chen. “Latent periodic process inference from single-cell RNA-seq data”. *Nature Communications* 11.1 (2020), p. 1441.

Additional publications

- [6 S] Ellie Haber, Shahul Alam, Nicholas Ho, Renming Liu, Evan Trop, **Shaoheng Liang**, Muyu Yang, Spencer Krieger, and Jian Ma. “Heimdall: A Modular Framework for Tokenization in Single-Cell Foundation Models”. *BioRxiv* (2025), pp. 2025–11.
- [7 J] Xianli Jiang, Nils-Petter Rudqvist, Bo Jiang, Shengbin Ye, Shan He, Qingnan Liang, Jinzhuang Dou, Michelle D Williams, Joe Dan Dunn, Jason M Johnson, et al. “Depletion of effector regulatory T cells associates with major response to induction dual immune checkpoint blockade”. *Cancer Discovery* 15.8 (2025), pp. 1569–1592.
- [8 J] Jincheng Han, Jiaqian Xu, Yonghong Liu, **Shaoheng Liang**, Kyle A LaBella, Deepavali Chakravarti, Denise J Spring, Yan Xia, and Ronald A DePinho. “Abstract IA20: Stromal-derived NRG1 enables oncogenic KRAS bypass in pancreas cancer”. *Cancer Research* 84.2_Supplement (2024), IA20–IA20.
- [9 J] Jiakai Hou, Yanjun Wei, Jing Zou, Roshni Jaffery, Long Sun, **Shaoheng Liang**, Ningbo Zheng, Ashley M Guerrero, Nicholas A Egan, Ritu Bohat, et al. “Integrated multi-omics analyses identify anti-viral host factors and pathways controlling SARS-CoV-2 infection”. *Nature Communications* 15.1 (2024), p. 109.
- [10 J] **Shaoheng Liang**, Jinzhuang Dou, Ramiz Iqbal, and Ken Chen. “Label-aware distance mitigates temporal and spatial variability for clustering and visualization of single-cell gene expression data”. *Communications Biology* 7.1 (2024), p. 326.
- [11 J] Huihui Fan, Feng Wang, Andy Zeng, Alex Murison, Katarzyna Tomczak, Dapeng Hao, Fatima Zahra Jelloul, Bofei Wang, Praveen Barroodia, **Shaoheng Liang**, et al. “Single-cell chromatin accessibility profiling of acute myeloid leukemia reveals heterogeneous lineage composition upon therapy-resistance”. *Communications Biology* 6.1 (2023), p. 765.
- [12 J] Jincheng Han, Jiaqian Xu, Yonghong Liu, **Shaoheng Liang**, Kyle A LaBella, Deepavali Chakravarti, Denise J Spring, Yan Xia, and Ronald A DePinho. “Stromal-derived NRG1 enables oncogenic KRAS bypass in pancreas cancer”. *Genes & Development* 37.17-18 (2023), pp. 818–828.
- [13 J] Li Li, Vakul Mohanty, Jinzhuang Dou, Yuefan Huang, Pinaki P Banerjee, Qi Miao, Jens G Lohr, Tushara Vijaykumar, Julia Frede, Birgit Knoechel, et al. “Loss of metabolic fitness drives tumor resistance after CAR-NK cell therapy and can be overcome by cytokine engineering”. *Science Advances* 9.30 (2023), eadd6997.
- [14 S] **Shaoheng Liang**, Xianli Jiang, Yulun Chiu, Haodong Xu, Kun Hee Kim, Gregory Lizée, and Ken Chen. “An interpretable ML model to characterize patient-specific HLA-I antigen presentation”. *BioRxiv* (2023).
- [15 J] Yukun Tan, Vakul Mohanty, **Shaoheng Liang**, Jinzhuang Dou, Jun Ma, Kun Hee Kim, Marc Jan Bonder, Xinghua Shi, Charles Lee, Zechen Chong, et al. “novoRNABreak: local assembly for novel splice junction and fusion transcript detection from RNA-seq data”. *Journal of Bioinformatics and Systems Biology: Open Access* 6.2 (2023), p. 74.
- [16 J] Li Li*, Muharrem Muftuoglu*, **Shaoheng Liang***, Mahesh Basyal, Jiangxing Lv, Mehmet Emin Akdogan, Ken Chen, Michael Andreeff, Christopher R Flowers, and Simrit Parmar. “In-depth analysis of SARS-CoV-2-specific T cells reveals diverse differentiation hierarchies in vaccinated individuals”. *JCI Insight* 7.7 (2022), e156559.
- [17 T] **Shaoheng Liang**. “Interpretable and Efficient Machine Learning in Cancer Biology”. PhD thesis. Rice University, 2022.

- [18] J] [Shaoheng Liang](#), Jason Willis, Jinzhuang Dou, Vakul Mohanty, Yuefan Huang, Eduardo Vilar, and Ken Chen. "Sensei: how many samples to tell a change in cell type abundance?" *BMC Bioinformatics* 23.1 (2022), p. 2.
- [19] J] Maria E Monberg, Heather Geiger, Jaewon J Lee, Roshan Sharma, Alexander Semaan, Vincent Bernard, Justin Wong, Fang Wang, [Shaoheng Liang](#), Daniel B Swartzlander, et al. "Occult polyclonality of preclinical pancreatic cancer models drives in vitro evolution". *Nature Communications* 13.1 (2022), p. 3652.
- [20] J] Hussein A Abbas, Vakul Mohanty, Ruiping Wang, Yuefan Huang, [Shaoheng Liang](#), Feng Wang, Jianhua Zhang, Yihua Qiu, Chenyue W Hu, Amina A Qutub, et al. "Decoupling lineage-associated genes in acute myeloid leukemia reveals inflammatory and metabolic signatures associated with outcomes". *Frontiers in Oncology* 11 (2021), p. 705627.
- [21] J] [Shaoheng Liang](#), Qingnan Liang, Rui Chen, and Ken Chen. "Stratified test accurately identifies differentially expressed genes under batch effects in single-cell data". *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 18.6 (2021), pp. 2072–2079.
- [22] J] Muharrem Muftuoglu, Li Li, Mahesh Basyal, [Shaoheng Liang](#), Carissa J Ball, Ken Chen, Prasanna Kumar, Arnaud Lesegretain, Naval Daver, and Michael Andreeff. "Single-cell proteomics identifies leukemia landscape associated with clinical outcomes in R/R AML treated with MDM2i (milademetan) and FLT3i (quizartinib): putative role of CD68 and diversity index". *Blood* 138 (2021), p. 3443.
- [23] J] Muharrem Muftuoglu, Li Li, [Shaoheng Liang](#), Duncan Mak, Angelique J Lin, Junxiang Fang, Jared K Burks, Ken Chen, and Michael Andreeff. "Extended live-cell barcoding approach for multiplexed mass cytometry". *Scientific Reports* 11.1 (2021), p. 12388.
- [24] J] Fang Wang, Qihan Wang, Vakul Mohanty, [Shaoheng Liang](#), Jinzhuang Dou, Jincheng Han, Darlan Conterno Minussi, Ruli Gao, Li Ding, Nicholas Navin, et al. "MEDALT: single-cell copy number lineage tracing enabling gene discovery". *Genome Biology* 22.1 (2021), p. 70.
- [25] C] [Shaoheng Liang](#), Qingnan Liang, Rui Chen, and Ken Chen. "Stratified test alleviates batch effects in single-cell data". *International Conference on Algorithms for Computational Biology*. 2020, pp. 167–177.
- [26] J] Fang Wang, Qihan Wang, Jincheng Han, Vakul Mohanty, [Shaoheng Liang](#), Darlan Conterno Minussi, Ruli Gao, Li Ding, Nicholas Navin, and Ken Chen. "Medatree enabling single cell copy number lineage tracing and functional discovery". *Cancer Research* 80.16_Supplement (2020), p. 4424.
- [27] J] Fang Wang, Qihan Wang, Vakul Mohanty, [Shaoheng Liang](#), Jinzhuang Dou, Jincheng Han, Darlan Minussi, Ruli Gao, Li Ding, Nicholas Navin, et al. "Abstract PO-017: Single-cell copy number heterogeneity tracing enabling cancer gene discovery". *Cancer Research* 80.21_Supplement (2020), PO-017.
- [28] J] Fang Wang, Qihan Wang, Vakul Mohanty, [Shaoheng Liang](#), Jinzhuang Dou, Jincheng Han, Darlan Conterno Minussi, Ruli Gao, Li Ding, Nicholas Navin, et al. "Minimal Event Distance Aneuploidy Lineage Tree (MEDALT) inference based on single cell copy number profile" (2020).
- [29] J] Qingnan Liang, Rachayata Dharmat, Leah Owen, Akbar Shakoor, Yumei Li, Sangbae Kim, Albert Vitale, Ivana Kim, Denise Morgan, [Shaoheng Liang](#), et al. "Single-nuclei RNA-seq on human retinal tissue provides improved transcriptome profiling". *Nature Communications* 10.1 (2019), p. 5743.
- [30] J] Fang Wang, [Shaoheng Liang](#), Tapsi Kumar, Nicholas Navin, and Ken Chen. "SCMarker: ab initio marker selection for single cell transcriptome profiling". *PLoS Computational Biology* 15.10 (2019), e1007445.
- [31] C] Jiansheng Chen, Gaocheng Bai, [Shaoheng Liang](#), and Zhengqin Li. "Automatic image cropping: A computational complexity study". *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 507–515.
- [32] C] [Shaoheng Liang](#), Jiansheng Chen, Zhengqin Li, and Gaocheng Bai. "Linear time symmetric axis search based on palindrome detection". *2016 IEEE International Conference on Image Processing (ICIP)*. 2016, pp. 1799–1803.

INVITED TALKS

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|---|----------------------|
| • Large Language Models in Computational Biology – A Primer | 2023 |
| C3.ai Digital Transformation Institute Generative AI conference | Champaign–Urbana, IL |
| • Manifold Preserving Feature Selection | 2021 |
| 2nd Annual GCC Single Cell Omics Cluster Symposium | Houston, TX |
| • Manifold Preserving Feature Selection | 2021 |
| Cancer Target Discovery and Development (CTD2) Network Oregon Health Science University (OSHU) site visit | Virtural |
| • Batch effect, Batch Stratified Test, and Batch Orthogonal Analysis | 2019 |
| Texas Single-Cell (TXSC) Seminar Series | Houston, TX |
| • Latent Variable Model Helps Delineate Cell States | 2019 |
| Texas Single-Cell (TXSC) Seminar Series | Houston, TX |

CONFERENCE TALKS

- **STEAMBOAT: Attention-based multiscale delineation of cellular interactions in tissues** 2025
International Conference on Research in Computational Molecular Biology (RECOMB) Seoul, South Korea
- **Manifold Preserving Feature Selection** 2021
CZI Seed Networks Computational Biology Meeting Virtual
- **Latent periodic process inference from single-cell RNA-seq data** 2020
Intelligent Systems for Molecular Biology (ISMB) Virtual
- **Stratified Test Alleviates Batch Effects in Single-Cell Data** 2020 2021
International Conference on Algorithms for Computational Biology ~~Missoula, MT~~ Virtual
- **Latent Variable Model Helps Delineate Cell States** 2018
Rice University Data Science Conference Houston, TX

TEACHING

- **Local alignment and BWT** | Guest lecture, CMU 02-710/02-510 Spring 26 2026
- **Introduction to spatial omics** | Guest lecture, CMU 02-710/02-510 Spring 25 2025
- **Multimomics integration and feature selection** | Instructor, Gulf Coast Consortia (GCC) Single-Cell Omics Workshop 2026
- **Introduction to Phylogenetics** | Teaching demos, Practicum in Teaching 2021
- **Recitation classes** | Teaching assistant, Statistical Machine Learning 2018 – 2020
- **Recitation classes** | Teaching assistant, Statistical Models and Algorithms for Data Science 2018 – 2019

RESEARCH MENTORSHIP

- Sizhe Liu (now at MIT) | Visiting pre-doctoral student 2025 – 2026
- Yufan Cao (now at MIT) | Visiting undergraduate student 2023 – 2024
- Guanghan Wang (now at Cornell Tech) | Visiting undergraduate student 2023
- Alice Feng (now at Harvard) | High school student 2021

MEDIA COVERAGE

- **New Single-Cell Algorithm Finds Gene Panels Used to ID Rare Cells** | GenomeWeb 2021